

Classifying positive energy modules in conformal field theory

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Motivation

Bottom-up approach to constructing conformal field theories:

- Pick a chiral algebra (vertex operator algebra)
- Classify modules in a suitable category
- Compute characters and their modular properties
- Find modular invariants
- Compute fusion products of modules
- Compute correlation functions

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Classifying positive energy modules

vertex operator algebras

ideals and quotients

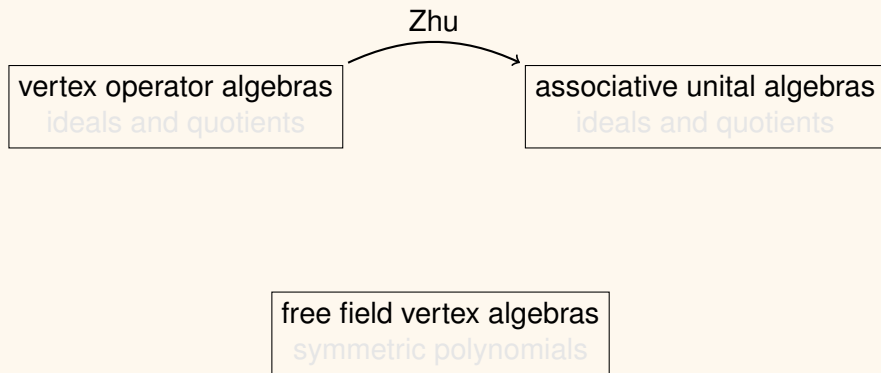
associative unital algebras

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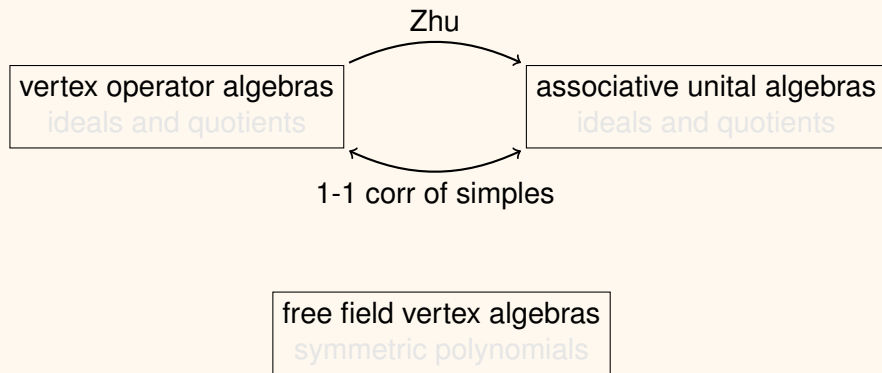
free field vertex algebras

symmetric polynomials

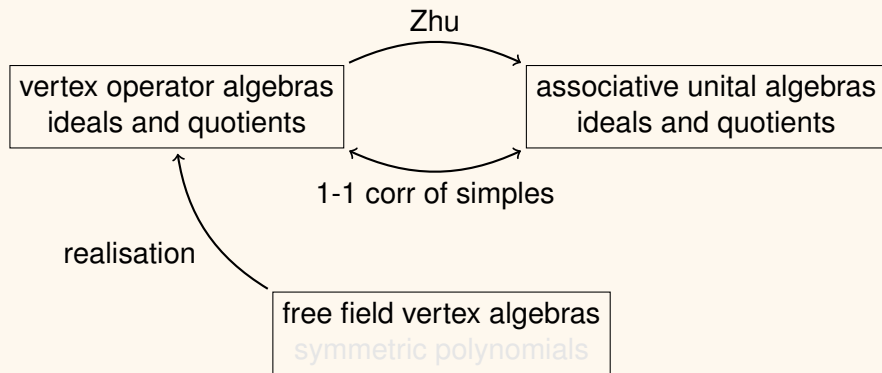
Classifying positive energy modules



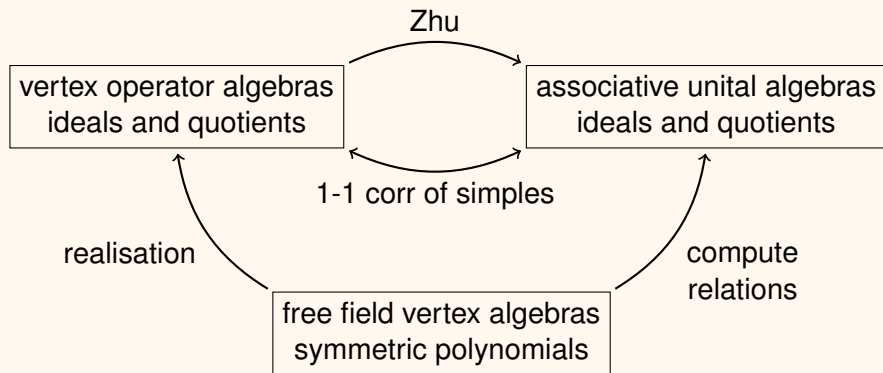
Classifying positive energy modules



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Sketch of Zhu's algebra

Let $\mathcal{L}$ be the Lie algebra of modes of a VOA V and let $U(\mathcal{L})_0$ be the centraliser of L_0 in $U(\mathcal{L})$. Define the zero mode map

$$V \rightarrow U(\mathcal{L})_0, A \mapsto A_0.$$

Let $\pi : U(\mathcal{L})_0 \rightarrow U(\mathcal{L})_0$ be the projection, where the kernel is spanned by Poincaré-Birkhoff-Witt basis elements with non-zero modes. Zhu's algebra is the image of

$$V \ni A \mapsto [A] = \pi(A_0).$$

The associative product is given by

$$[A] * [B] = \pi(A_0 B_0)$$

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Universal Virasoro vertex operator algebra

The universal Virasoro vertex operator algebra $\text{Vir}_c, c \in \mathbb{C}$:

- C) 1 generating field $T = \sum_{n \in \mathbb{Z}} L_n z^{-n-2}$
- C) OPE relation: $T(z)T(w) = \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w} + \text{reg}$
- P) Commutation relations: $[L_m, L_n] = (m-n)L_{m+n} + \delta_{m+n,0} \frac{c}{12}(m^3 - m)$
- C) Basis: $:\partial^{n_1} T(z) \cdots \partial^{n_\ell} T(z):, n_1 \geq \cdots \geq n_\ell \geq 0$.
- P) Zhu algebra: $\mathbb{C}[T]$

Virasoro minimal model vertex operator algebra

- The universal Virasoro vertex operator algebra has a proper ideal if and only if there exist $p, q \geq 2$, $\gcd(p, q) = 1$ so that

$$c = 1 - 6 \frac{(p - q)^2}{pq} = c_{p,q},$$

- The ideal is generated by a singular vector of conformal weight $(p - 1)(q - 1)$.
- The Virasoro minimal model vertex operator algebra $M(p, q)$ is the quotient by this ideal.

As a vertex operator algebra:

- 1 generating field $T = \sum_{n \in \mathbb{Z}} L_n z^{-n-2}$
- Same OPE and commutation relations.
- Spanning set: $:\partial^{n_1} T(z) \cdots \partial^{n_\ell} T(z):$, $n_1 \geq \cdots \geq n_\ell \geq 0$.
- Zhu algebra: $\mathbb{C}[T] / \langle \text{image of singular vector} \rangle$

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Module classification: Yang-Lee model, $c_{2,5} = -22/5$

- Ideal generated by singular vector χ

$$\chi = (L_{-2}^2 - \frac{3}{5}L_{-4})\Omega$$

$$\chi(z) = :T(z)^2: - \frac{3}{10}\partial^2 T(z) = \sum_{n \in \mathbb{Z}} \chi_n z^{-n-4}$$

- Evaluate χ_0 on a candidate Virasoro highest weight vector Ω_h .

$$0 = \chi_0 \Omega_h = (L_0^2 + \frac{1}{5}L_0)\Omega_h = h(h + \frac{1}{5})\Omega_h$$

Image of χ in universal Zhu algebra, $\mathbb{C}[T]$, is $T(T + 1/5)$.

- Minimal model Zhu algebra: $\mathbb{C}[T] / \langle T(T + 1/5) \rangle$.
- Two inequivalent simple modules. Highest weights 0 and $-\frac{1}{5}$.

General classification scheme

- Zhu algebras for universal vertex operator algebras are easy.
- Finding singular vectors is very hard in general.
- Realise universal vertex operator algebras as subalgebras of free field algebras.
- Free field algebras admit screening operators which can be used to construct singular vectors.
- These singular vectors allow one to find presentations of the quotient Zhu algebras.

Virasoro minimal models

- Central charge

$$c_{p,q} = 1 - 6 \frac{(p-q)^2}{pq}$$

- Image of singular vector in Zhu algebra $\mathbb{C}[T]$:

$$f(T) = \prod_{\substack{r=1 \\ (r,s) \sim (p-r,q-s)}}^{p-1} \prod_{s=1}^{q-1} (T - h_{r,s}), \quad h_{r,s} = \frac{(ps - qr)^2 - (p - q)^2}{4pq}.$$

- Minimal model Zhu algebra: $\mathbb{C}[T] / \langle f(T) \rangle$

$\mathfrak{sl}(2)$ minimal models

C) Generated by standard basis of $\mathfrak{sl}(2)$: e, h, f

C) OPE relations: For $x, y \in \mathfrak{sl}(2)$

$$x(z)y(w) \sim \frac{\kappa(x, y)k}{(z-w)^2} + \frac{[x, y](w)}{z-w}, \quad k \neq -2.$$

P) There exists an ideal if and only if there exist integers $u \geq 2, v \geq 1$, $\gcd(u, v) = 1$ for which $k + 2 = \frac{u}{v}$.

P) The ideal is generated by a singular vector of conformal weight $(u-1)v$ and $\mathfrak{sl}(2)$ weight $2(u-1)$.

P) Zhu algebra: $U(\mathfrak{sl}(2))$

P) Quadratic Casimir: $Q = h^2/2 + ef + fe$. Image of singular vector

$$e^{u-1} \prod_{i=1}^{u-1} \prod_{\substack{j=1 \\ (i,j) \sim (u-i, v-j)}}^{v-1} (Q - q_{i,j}), \quad q_{i,j} = u \frac{(vi - uj)^2 - v^2}{v^2}$$

$\mathfrak{sl}(2)$ minimal models. Simple module classification

Image of singular vector

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$(i,j) \sim (u-i, v-j)$

- Simple positive energy affine $\mathfrak{sl}(2)$ modules are quotients of inductions from simple $\mathfrak{sl}(2)$ weight modules.
- Simple $\mathfrak{sl}(2)$ weight modules are classified by the following list:
 - $\mathcal{F}_\lambda$, $\lambda \in \mathbb{Z}_{\geq 0}$: $\lambda + 1$ dimensional module of highest weight $\lambda\alpha$.
 - $\mathcal{V}_\lambda^+$, $\lambda \in \mathbb{C} \setminus \mathbb{Z}_{\geq 0}$: Infinite-dimensional highest weight module of weight $\lambda\alpha$.
 - $\mathcal{V}_\lambda^-$, $\lambda \in \mathbb{C} \setminus \mathbb{Z}_{\leq 0}$: Infinite-dimensional lowest weight module of weight $\lambda\alpha$.
 - $\mathcal{D}_{\Lambda,q}$, $q \in \mathbb{C}$, $\Lambda \in \mathbb{C}/2\mathbb{Z}$ and $q \neq \frac{1}{2}\lambda(\lambda + 2)$ for all $\lambda \in \Lambda$: Dense module, $Q = q \cdot \text{id}$.

$\mathfrak{sl}(2)$ minimal models. Simple module classification

Image of singular vector

$$e^{u-1} \prod_{i=1}^{u-1} \prod_{j=1}^{v-1} (Q - q_{i,j}), \quad q_{i,j} = u \frac{(vi - uj)^2 - v^2}{v^2}$$

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- Simple $\mathfrak{sl}(2)$ weight modules are classified by the following list:
 - $\mathcal{F}_\lambda$, $\lambda \in \mathbb{Z}_{\geq 0}$: $\lambda + 1$ dimensional module of highest weight $\lambda\alpha$. Must have $\lambda \leq u - 1$
 - $\mathcal{V}_\lambda^+$, $\lambda \in \mathbb{C} \setminus \mathbb{Z}_{\geq 0}$: Infinite-dimensional highest weight module of weight $\lambda\alpha$.
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For the infinite-dimensional modules we must have $Q = q_{i,j} \cdot \text{id}$

Free field realisation of the universal Virasoro VOA

Free boson or Heisenberg vertex operator algebra

- 1 generator a : $a(z)a(w) \sim \frac{1}{(z-w)^2}$.
- $T(z) = \frac{1}{2} : a(z)^2 : + \frac{\alpha_0}{2} \partial a(z)$, $c = 1 - 3\alpha_0^2$, $\alpha_0 \in \mathbb{C}$.
- For α_+, α_- , such that $\alpha_+ \alpha_- = -2$ and $\alpha_+ + \alpha_- = \alpha_0$,

$$S_{\pm}(w) = e^{\alpha_{\pm} \hat{a}} w^{\alpha_{\pm} \alpha_0} \prod_{m \geq 1} \exp \left(\alpha_{\pm} \frac{a_{-m} w^m}{m} \right) \exp \left(-\alpha_{\pm} \frac{a_m w^{-m}}{m} \right)$$

is a screening field: $Q_{\pm} = \text{Res}_{w=0} [S_{\pm}(w)]$ commutes with $T(z)$.

- $Q_{\pm}$ defines a map between Fock spaces $F_{\lambda} \rightarrow F_{\lambda+\alpha_{\pm}}$ whenever the residue is well defined.
- Since $Q_{\pm}$ commutes with $T(z)$, the image of the highest weight vector of F_{λ} will be **singular** in $F_{\lambda+\alpha_{\pm}}$.

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The singular vector at minimal model central charges

The vacuum singular vector can only be constructed by composing multiple screening operators.

Let $p, q \geq 2$, $\gcd(p, q) = 1$ and

$$\alpha_+ = \sqrt{\frac{2q}{p}}, \quad \alpha_- = -\sqrt{\frac{2p}{q}}, \quad \alpha_0 = \alpha_+ + \alpha_-, \quad c = 1 - 3\alpha_0^2 = 1 - 6\frac{(p-q)^2}{pq}.$$

The vacuum singular vector of the universal Virasoro VOA is

$$\chi = Q_+^{[p-1]}|(1-p)\alpha_+\rangle \propto Q_-^{[q-1]}|(1-q)\alpha_-\rangle.$$

The minimal model conformal weights are the zeros of the action of the zero mode of χ on highest weight vectors.

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Composing screening operators

To construct the singular vector we must compose screening fields.

$$S_+(z_1) \cdots S_+(z_r) = e^{r\alpha_+ \hat{a}} \prod_{i \neq j} (z_i - z_j)^{\alpha_+^2/2} \prod_{i=1}^r z_i^{\alpha_+ + a_0} \\ \prod_{m \geq 1} \exp \left(\alpha_+ \frac{a_{-m}}{m} \sum_i z_i^m \right) \exp \left(-\alpha_+ \frac{a_m}{m} \sum_i z_i^{-m} \right)$$

Proposition: If and only if $\mu = \frac{1-r}{2}\alpha_+ - \frac{k}{2}\alpha_-$, $k \in \mathbb{Z}$, there exists a unique choice of a cycle Γ such that

$$Q_+^{[r]} = \int_{\Gamma} S_+(z_1) \cdots S_+(z_r) dz_1 \cdots dz_r : F_{\frac{1-r}{2}\alpha_+ - \frac{k}{2}\alpha_-} \rightarrow F_{\frac{1+r}{2}\alpha_+ - \frac{k}{2}\alpha_-}$$

is a non-trivial Virasoro homomorphism.

- if $k \geq 0$ then $Q_+^{[r]} | \frac{1-r}{2}\alpha_+ - \frac{k}{2}\alpha_- \rangle = 0$.
- if $k < 0$ then $Q_+^{[r]} | \frac{1-r}{2}\alpha_+ - \frac{k}{2}\alpha_- \rangle \neq 0$.

Appearance of symmetric polynomials

Action of $Q_+^{[r]}$ on $F_{\frac{1-r}{2}\alpha_+ - \frac{k}{2}\alpha_-}$:

$$Q_+^{[r]} = e^{r\alpha_+ \hat{a}} \int_{\Gamma} \prod_{i \neq j} \left(1 - \frac{z_i}{z_j}\right)^{\alpha_+^2/2} \prod_{i=1}^r z_i^{k+1} \\ \times \prod_{m \geq 1} \exp\left(\alpha_+ \frac{a_{-m}}{m} \sum_i z_i^m\right) \exp\left(-\alpha_+ \frac{a_m}{m} \sum_i z_i^{-m}\right) \frac{dz_1 \cdots dz_r}{z_1 \cdots z_r}$$

This integral defines a well known inner product on the ring of symmetric polynomials $\mathbb{C}[z_1, \dots, z_r]^{\mathfrak{S}_r}$:

$$\langle f, g \rangle_r^t = \int_{\Gamma} \prod_{i \neq j} \left(1 - \frac{z_i}{z_j}\right)^{1/t} f(z_1^{-1}, \dots, z_r^{-1}) g(z_1, \dots, z_r) \frac{dz_1 \cdots dz_r}{z_1 \cdots z_r}$$

This inner product is symmetric and it vanishes if $\deg(f) \neq \deg(g)$.

Symmetric polynomials

Ring of symmetric polynomials in r variables: $\Lambda_r = \mathbb{C}[z_1, \dots, z_r]^{\mathfrak{S}_r}$

Many well known generators:

- Power sums

$$\mathbf{p}_k(z) = \sum_i z_i^k$$

- Elementary symmetric polynomials

$$\mathbf{e}_k(z) = \sum_{1 \leq i_1 < \dots < i_k} z_{i_1} \cdots z_{i_k}$$

$$\Lambda_r = \mathbb{C}[\mathbf{p}_1, \dots, \mathbf{p}_r] = \mathbb{C}[\mathbf{e}_1, \dots, \mathbf{e}_r]$$

Symmetric polynomials

Many well known bases. Generally indexed by partitions

$$\lambda = [\lambda_1, \lambda_2, \dots].$$

- Power sum basis: $p_\lambda = p_{\lambda_1} p_{\lambda_2} \cdots$
- Elementary symmetric basis: $e_\lambda = e_{\lambda_1} e_{\lambda_2} \cdots$
- Monomial symmetric polynomials:

$$m_\lambda(z) = \sum_{\sigma} z_1^{\lambda_{\sigma(1)}} z_2^{\lambda_{\sigma(2)}} \cdots,$$

where the σ are all distinct permutations of $\lambda = [\lambda_1, \lambda_2, \dots]$.

$$m_{[2,2]}(z_1, z_2) = z_1^2 z_2^2, \quad m_{[2,2]}(z_1, z_2, z_3) = z_1^2 z_2^2 + z_1^2 z_3^2 + z_2^2 z_3^2.$$

$$\Lambda_r = \bigoplus_{\lambda, \lambda_1 \leq r} \mathbb{C} p_\lambda = \bigoplus_{\lambda, \lambda_1 \leq r} \mathbb{C} e_\lambda = \bigoplus_{\lambda, \ell(\lambda) \leq r} \mathbb{C} m_\lambda$$

Jack symmetric polynomials

Jack symmetric polynomials $P_\lambda^t(z)$ are uniquely characterised by

- 1 Upper triangular: $P_\lambda^t(z) = m_\lambda(z) + \sum_{\nu < \lambda} c_{\nu,\lambda}(t) m_\nu(z)$
- 2 Orthogonal: $\langle P_\lambda^t, P_\nu^t \rangle_r^t = 0$, if $\lambda \neq \nu$.

Well studied:

- $\langle P_\lambda^t(z), P_\lambda^t(z) \rangle_r^t = a_\lambda(t)$
- $\prod_{i,j} (1 - y_i z_j)^{-1/t} = \prod_{m \geq 1} \exp\left(\frac{1}{t} \frac{p_m(y) p_m(z)}{m}\right) = \sum_\nu b_\nu(t) P_\nu^t(y) P_\nu^t(z)$
- $\prod_i (1 + z_i)^{X/t} = \sum_\lambda d_\lambda(X, t) P_\lambda^t(z)$

Where $a_\lambda(t)$, $b_\lambda(t)$, $c_{\nu,\lambda}(t)$, $d_\lambda(X, t)$ are rational functions in t . Explicit formulae known.

Examples:

- $P_{[1,\dots,1]}^t(z) = e_{[1,\dots,1]}(z)$
- $P_{[s,\dots,s]}^t(z_1, \dots, z_r) = \prod_{i=1}^r z_i^s, \ell([s, \dots, s]) = r$

Determining conformal weights from the zero mode

We want to evaluate the zero mode of the singular vector

$$Q_+^{[p-1]}|(1-p)\alpha_+\rangle = \int_{\Gamma} S_+(z_1) \cdots S_+(z_{p-1})|(1-p)\alpha_+\rangle dz_1 \cdots dz_{p-1}.$$

The corresponding field is

$$\begin{aligned} Q_+^{[p-1]}V_{(1-p)\alpha_+}(w) \\ = \int_{\Gamma} S_+(z_1 + w) \cdots S_+(z_{p-1} + w)V_{(1-p)\alpha_+}(w) dz_1 \cdots dz_{p-1}. \end{aligned}$$

The easiest way to evaluate such a zero mode is to evaluate matrix elements

$$\langle \tau | Q_+^{[p-1]}V_{(1-p)\alpha_+}(w) | \tau \rangle,$$

where $\tau \in \mathbb{C}$ and $\langle \tau | a_{-n} = 0, n \geq 1$.

Determining conformal weights from the zero mode

$$\langle \tau | Q_+^{[p-1]} V_{(1-p)\alpha_+}(w) | \tau \rangle = \left\langle P_{[(q-1)p-1]}^{p/q}(z), \prod_{i=1}^{p-1} \left(1 + \frac{z_i}{w}\right)^{\alpha_+ \tau} \right\rangle_{p-1}^{p/q}$$

There are known formulae for expanding

$$\prod_{i=1}^{p-1} \left(1 + \frac{z_i}{w}\right)^{\alpha_+ \tau} = \sum_{\nu} g_{\nu}(\alpha_+ \tau) P_{\nu}^{p/q}(z) w^{-|\nu|}.$$

Determining conformal weights from the zero mode

$$\langle \tau | Q_+^{[p-1]} V_{(1-p)\alpha_+}(w) | \tau \rangle \propto g_{[(q-1)^{p-1}]}(\alpha_+ \tau) \propto \prod_{r,s} (h_\tau - h_{r,s}),$$

where $1 \leq r \leq p-1$, $1 \leq s \leq q-1$, $(r, s) \simeq (p-r, q-s)$ and

$$h_\tau = \frac{\tau}{2}(\tau - \alpha_0), \quad h_{r,s} = \frac{(ps - qr)^2 - (p - q)^2}{4pq}.$$

Theorem

Let $f(T) = \prod_{r,s} (T - h_{r,s})$.

Minimal model Zhu algebra: $\frac{\mathbb{C}[T]}{\langle f(T) \rangle}$

The minimal model highest (conformal) weights are the zeros of $f(T)$.

Other algebras: $\mathfrak{sl}(2)$

Non-simple levels: $k + 2 = \frac{u}{v}$, $u \geq 2$, $v \geq 1$, $\gcd(u, v) = 1$.

Matrix element to be evaluated:

$$\left\langle \mathbf{P}_{[(v-1)^{u-1}]}^{u/v}, \prod_{i=1}^{u-1} (1 + z_i)^{-\sqrt{\frac{2v}{u}} \tau - 1} \right\rangle^{u/v} \propto \prod_{i=1}^{u-1} \prod_{\substack{j=1 \\ (i,j) \sim (u-i, v-j)}}^{v-1} (q_\tau - q_{i,j}),$$

τ free boson weight, q_τ eigenvalue of quadratic Casimir.

Image of singular vector in Zhu algebra $U(\mathfrak{sl}(2))$:

$$e^{u-1} \prod_{i=1}^{u-1} \prod_{\substack{j=1 \\ (i,j) \sim (u-i, v-j)}}^{v-1} (Q - q_{i,j})$$

Other algebras: $N = 1$ Virasoro

- Generated by 1 even and 1 odd field: T, G .
- OPE relations

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}$$

$$T(z)G(w) \sim \frac{3/2G(w)}{(z-w)^2} + \frac{\partial G(w)}{z-w}$$

$$G(z)G(w) \sim \frac{2c/3}{(z-w)^3} + \frac{2T(w)}{z-w}$$

- Non-simple if and only if there exist $p, q \geq 2$, $p - q \in 2\mathbb{Z}$, $\gcd(p, \frac{p-q}{2}) = 1$, so that $c = \frac{3}{2} - 3\frac{(p-q)^2}{pq}$.
- Singular vector at level $(p-1)(q-1)/2$
- NS Zhu algebra: $\mathbb{C}[T]$
- R Zhu algebra: $\mathbb{C}[G] = \mathbb{C}[T] \oplus G\mathbb{C}[T]$, $G^2 = T - c/24$

Other algebras: $N = 1$ Virasoro

To find image of singular vector evaluate the following matrix elements:

$$\text{NS : } \left\langle \mathbf{P}_{\delta}^{-3}, \prod_{i=1}^{p-1} (1 + z_i) \sqrt{\frac{q}{p}} \tau \right\rangle^{2p/(q-p)} = \prod_{\substack{r=1 \\ (r,s) \sim (p-r, q-s) \\ r+s \in 2\mathbb{Z}}}^{p-1} \prod_{s=1}^{q-1} (h_{\tau} - h_{r,s}^{(\text{NS})})$$

$$\text{R : } \left\langle \mathbf{P}_{\delta}^{-3}, \prod_{i=1}^{p-1} (1 + z_i) \sqrt{\frac{q}{p}} \tau^{-\frac{1}{2}} \right\rangle^{2p/(q-p)} = \prod_{\substack{r=1 \\ (r,s) \sim (p-r, q-s) \\ r+s \in 2\mathbb{Z}+1}}^{p-1} \prod_{s=1}^{q-1} (h_{\tau} - h_{r,s}^{(\text{R})})$$

τ free boson weight; h_{τ} conformal weight.

- NS Zhu algebra: $\mathbb{C}[T]$
- Image of singular vector in NS sector: $\prod_{r,s} (T - h_{r,s}^{(\text{NS})})$
- R Zhu algebra: $\mathbb{C}[T] \oplus G\mathbb{C}[T]$
- Image of singular vector in R sector:
 - $\prod_{r,s} (T - h_{r,s}^{(\text{R})})$, if p, q odd,
 - $G \prod_{r,s} (T - h_{r,s}^{(\text{R})})$, if p, q even.

Other algebras: $N = 1$ Virasoro

To find image of singular vector evaluate the following matrix elements:

$$\text{NS : } \left\langle \mathbf{P}_{\delta}^{-3}, \prod_{i=1}^{p-1} (1 + z_i) \sqrt{\frac{q}{p}} \tau \right\rangle^{2p/(q-p)} = \prod_{\substack{r=1 \\ (r,s) \sim (p-r, q-s) \\ r+s \in 2\mathbb{Z}}}^{p-1} \prod_{s=1}^{q-1} (h_{\tau} - h_{r,s}^{(\text{NS})})$$

$$\text{R : } \left\langle \mathbf{P}_{\delta}^{-3}, \prod_{i=1}^{p-1} (1 + z_i) \sqrt{\frac{q}{p}} \tau^{-\frac{1}{2}} \right\rangle^{2p/(q-p)} = \prod_{\substack{r=1 \\ (r,s) \sim (p-r, q-s) \\ r+s \in 2\mathbb{Z}+1}}^{p-1} \prod_{s=1}^{q-1} (h_{\tau} - h_{r,s}^{(\text{R})})$$

τ free boson weight; h_{τ} conformal weight.

- NS Zhu algebra: $\mathbb{C}[T]$
- Image of singular vector in NS sector: $\prod_{r,s} (T - h_{r,s}^{(\text{NS})})$
- R Zhu algebra: $\mathbb{C}[T] \oplus G\mathbb{C}[T]$
- Image of singular vector in R sector:
 - $\prod_{r,s} (T - h_{r,s}^{(\text{R})})$, if p, q odd,
 - $G \prod_{r,s} (T - h_{r,s}^{(\text{R})})$, if p, q even.

Other algebras: universal $\mathfrak{osp}(1|2)$ vertex operator algebra

- Generated by basis of $\mathfrak{osp}(1|2)$: even e, h, f , odd x, y

$$[e, x] = [f, y] = 0, \quad [e, y] = -x, \quad [f, x] = -y, \quad [h, x] = x, \quad [h, y] = -y \\ \{x, x\} = 2e, \quad \{x, y\} = h, \quad \{y, y\} = -2f.$$

- OPE relations: for $g, h \in \mathfrak{osp}(1|2)$

$$g(z)h(w) \sim \frac{\kappa(g, h)k}{(z-w)^2} + \frac{[g, h](w)}{z-w}, \quad k \neq -\frac{3}{2}.$$

- Non-simple levels: $2k+3 = \frac{u}{v}$, $u \geq 2$, $v \geq 1$, $u-v \in 2\mathbb{Z}$, $\gcd(u, \frac{u-v}{2}) = 1$.
- Singular vector at conformal weight $(u-1)v/2$ and $\mathfrak{osp}(1|2)$ weight $u-1$.
- NS Zhu algebra: $U(\mathfrak{osp}(1|2))$.
- R Zhu algebra: $U(\mathfrak{sl}(2))$.

Other algebras: universal $\mathfrak{osp}(1|2)$ vertex operator algebra

To find the image of the singular vector evaluate the matrix elements

$$\text{NS} : \left\langle P_{\delta}^{-3}, \prod_{i=1}^{u-1} (1 + z_i)^{-\sqrt{\frac{v}{u}}/\tau - 1} \right\rangle^{2u/(v-u)}$$

$$\text{R} : \left\langle P_{\delta}^{-3}, \prod_{i=1}^{u-1} (1 + z_i)^{\sqrt{\frac{v}{u}}/\tau - \frac{1}{2}} \right\rangle^{2u/(v-u)}$$

Image of singular vector:

- u, v odd:

$$\text{NS}: e^{(u-1)/2} \prod_{i=1}^{u-1} \prod_{\substack{j=1 \\ i+j \in 2\mathbb{Z}+1}}^{v-1} (\Sigma - s_{i,j})$$

$$\text{R}: e^{(u-1)/2} \prod_{i=1}^{u-1} \prod_{\substack{j=1 \\ i+j \in 2\mathbb{Z} \\ (i,j) \sim (u-i, v-j)}}^{v-1} (Q - q_{i,j}) .$$

Other algebras: universal $\mathfrak{osp}(1|2)$ vertex operator algebra

To find the image of the singular vector evaluate the matrix elements

$$\text{NS} : \left\langle \mathbf{P}_{\delta}^{-3}, \prod_{i=1}^{u-1} (1 + z_i)^{-\sqrt{\frac{v}{u}}/\tau - 1} \right\rangle^{2u/(v-u)}$$

$$\text{R} : \left\langle \mathbf{P}_{\delta}^{-3}, \prod_{i=1}^{u-1} (1 + z_i)^{\sqrt{\frac{v}{u}}/\tau - \frac{1}{2}} \right\rangle^{2u/(v-u)}$$

Image of singular vector:

- u, v even:

$$\text{NS}: e^{(u-2)/2} x \prod_{i=1}^{u-1} \prod_{\substack{j=1 \\ i+j \in 2\mathbb{Z}+1}}^{v-1} (\Sigma - s_{i,j})$$

$$\text{R}: e^{(u-2)/2} \prod_{i=1}^{u-1} \prod_{\substack{j=1 \\ i+j \in 2\mathbb{Z}}}^{v-1} (Q - q_{i,j}) .$$

$(i,j) \sim (u-i, v-j)$

The End

Thank you for coming!